

MATH 345 Skeleton Notes

Unit 2: Anatomy of a linear map

Matrix invertibility and linear systems

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Matrix invertibility and linear systems

In this section we finish the square-matrix version of the Unit 2 story. A square matrix is invertible exactly when it “does not forget any nonzero input direction” and “every output is reached once”. The inverse matrix gives the inverse map when it exists. The determinant, a fundamental quantity in linear algebra that we only briefly discuss in this course, gives a single-number test for invertibility.

Definition: Invertibility of Matrices

An $n \times n$ matrix A is called **nonsingular**, or **invertible**, if there exists an $n \times n$ matrix B such that $AB = BA = I_n$. Such a B is called the **inverse** of A . If no such B exists, A is called **singular** or **noninvertible**.

Note

The inverse of a matrix mimics the reciprocal of a real number.

Activity: Testing whether two matrices are inverses (U2-LO5)

Let

$$A = \begin{bmatrix} 1 & 3 \\ 1 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 4 & -3 \\ -1 & 1 \end{bmatrix}.$$

Compute both AB and BA , and make a conclusion using the language of inverses.

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2 \quad \text{and} \quad BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

Therefore, B is the inverse of A .

Fact

The inverse of a matrix, if it exists, is unique. Therefore, we can write A^{-1} for the inverse of A .

Fact

For $n \times n$ matrices A and B , if $AB = I_n$, then $BA = I_n$.

Activity: Testing a matrix for invertibility (U2-LO5)

Does $A = \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$ have an inverse?

Let's suppose A had an inverse $B = \begin{bmatrix} x & y \\ z & w \end{bmatrix}$.

Then:

$$AB = I_2$$
$$\begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

This gives us the system of linear equations in four unknowns:

$$\begin{array}{rclcl} 2x & & + & z & = 1 \\ & 2y & & + & w = 0 \\ 0x & & + & 0z & = 0 \\ & 0y & & + & 0w = 1 \end{array}$$

The last equation states that $0 = 1$, which is impossible.

Therefore, A cannot have an inverse. This means A is singular.

Theorem

If both A and B are nonsingular $n \times n$ matrices, then the matrix AB is

nonsingular and its inverse is $(AB)^{-1} = B^{-1}A^{-1}$.

Why is this true?

It suffices to calculate that $(AB)(B^{-1}A^{-1}) = I_n$ and $(B^{-1}A^{-1})(AB) = I_n$.
Firstly,

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_nA^{-1} = AA^{-1} = I_n.$$

Similarly,

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}I_nB = B^{-1}B = I_n.$$

Fact: Follow Up Facts About Invertibility

- If $A_1, A_2, \dots, A_k$ are $n \times n$ invertible / nonsingular matrices, then $A_1A_2 \cdots A_k$ is invertible / nonsingular and

$$(A_1A_2 \cdots A_k)^{-1} = A_k^{-1} \cdots A_2^{-1}A_1^{-1}.$$

- If A is invertible/nonsingular, then A^{-1} is invertible/nonsingular and

$$(A^{-1})^{-1} = A.$$

- If A is invertible/nonsingular, then A^T is invertible/nonsingular and

$$(A^{-1})^T = (A^T)^{-1}.$$

Let $T = T_A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a matrix transformation induced by the $n \times n$ matrix A . What does invertibility imply about the geometry of the transformation T_A ?

Consider $T' = T_{A^{-1}}$. What happens when we compose them?

Activity: Composing a map and its inverse (U2-L05)

Compute the composition of T and T' in both orders.

Let's examine the composition of T and T' :

$$\begin{aligned}T \circ T'(\mathbf{x}) &= T_A(T_{A^{-1}}(\mathbf{x})) \\&= T_A(A^{-1}\mathbf{x}) \\&= A(A^{-1}\mathbf{x}) \\&= (AA^{-1})\mathbf{x} \\&= I_n\mathbf{x} \\&= \mathbf{x}\end{aligned}$$

Similarly, for the reverse composition:

$$\begin{aligned}T' \circ T(\mathbf{x}) &= T_{A^{-1}}(T_A(\mathbf{x})) \\&= T_{A^{-1}}(A\mathbf{x}) \\&= A^{-1}(A\mathbf{x}) \\&= (A^{-1}A)\mathbf{x} \\&= I_n\mathbf{x} \\&= \mathbf{x}\end{aligned}$$

Definition

Given a matrix transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$, if there exists a transformation $S : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $S(T(\mathbf{x})) = T(S(\mathbf{x})) = \mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$, we say that T has an inverse.

Theorem

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ denote the matrix transformation induced by an $n \times n$ matrix A . Then:

1. A is invertible if and only if T has an inverse.
2. In this case, T has exactly one inverse (which we denote as T^{-1}), and $T^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the transformation induced by the matrix A^{-1} .

In other words, $(T_A)^{-1} = T_{A^{-1}}$.

T and T^{-1} ‘reverse’ or ‘undo’ the action of each other.

Activity: Reversing a reflection (U2-L05)

Consider $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by reflection across the x -axis. Find the matrix representing this transformation, and find the inverse.

T is given by matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Since $A^2 = I_2$, A is its own inverse, so $T^{-1} = T$.

Theorem

The 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is invertible if and only if $ad - bc \neq 0$, in which case

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

We define the **determinant** of the matrix A to be the quantity $\det(A) = ad - bc$. Thus the matrix A is invertible if and only if its determinant is nonzero.

For now, $\det(A) = ad - bc$ is only the determinant of a 2×2 matrix. We will define determinants for larger square matrices later in this section.

Activity: Reading determinant checks in code (U2-LO3, U2-LO5, U2-LO7)

The following code computes determinants of two 2×2 matrices.

```
import numpy as np

A = np.array([
    [1., 1.],
    [0., 1.],
])

B = np.array([
    [1., 0.],
    [0., 0.],
])

detA = float(np.linalg.det(A))
detB = float(np.linalg.det(B))

detA, detB
```

Output:

```
(1.0, 0.0)
```

Activity: Reading determinant checks in code (U2-LO3, U2-LO5, U2-LO7)
(continued)

1. Which matrix is invertible?
2. Which matrix forgets a nonzero input direction?
3. How does the determinant tell you?

The output says

$$\det(A) = 1, \quad \det(B) = 0.$$

Matrix A is invertible because its determinant is nonzero. Matrix B is singular because its determinant is zero.

The matrix B forgets the nonzero input direction

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

since

$$B \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

For a 2×2 matrix, a nonzero determinant means

the matrix is invertible. A zero determinant means the matrix is singular and forgets some nonzero input direction.

Activity: Checking invertibility and finding inverses (U2-LO5)

Find the inverse of each matrix, if possible.

$$A = \begin{bmatrix} -2 & -3 \\ 4 & 6 \end{bmatrix}$$

$\det(A) = (-2)(6) - (-3)(4) = -12 - (-12) = 0$,
so A is not invertible.

Activity: Checking invertibility and finding inverses (U2-LO5)

Find the inverse of each matrix, if possible.

$$B = \begin{bmatrix} 1 & 6 \\ 2 & 3 \end{bmatrix}$$

$$\det(B) = (1)(3) - (6)(2) = 3 - 12 = -9 \neq 0,$$

so

$$B^{-1} = \frac{1}{-9} \begin{bmatrix} 3 & -6 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -1/3 & 2/3 \\ 2/9 & -1/9 \end{bmatrix}.$$

If A is an $n \times n$ matrix, then the linear system $A\mathbf{x} = \mathbf{b}$ is a system of n equations in n unknowns. Suppose A is nonsingular. How can we use A^{-1} to solve the system $A\mathbf{x} = \mathbf{b}$?

Activity: Solving a system with an inverse (U2-LO5)

Solve the system using the inverse of A .

Multiply both sides of $A\mathbf{x} = \mathbf{b}$ by A^{-1} on the left:

$$\begin{aligned}
A\mathbf{x} &= \mathbf{b} \\
A^{-1}(A\mathbf{x}) &= A^{-1}\mathbf{b} \\
(A^{-1}A)\mathbf{x} &= A^{-1}\mathbf{b} \\
I_n\mathbf{x} &= A^{-1}\mathbf{b} \\
\mathbf{x} &= A^{-1}\mathbf{b}
\end{aligned}$$

Therefore, when A is nonsingular, $\mathbf{x} = A^{-1}\mathbf{b}$ is the solution to $A\mathbf{x} = \mathbf{b}$.

Consequences:

- When A^{-1} exists, then $A\mathbf{x} = \mathbf{b}$ has a unique solution.
- If A is invertible/nonsingular, then the *only* solution to the homogeneous system $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$.
- Put differently, an invertible square matrix forgets no nonzero input direction.

Activity: One inverse, several right-hand sides (U2-L05)

Suppose that

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Use this information to calculate the solution to the following systems:

$$A\mathbf{x} = \mathbf{b}$$

Since $\det(A) = (1)(4) - (2)(3) = -2 \neq 0$, the 2×2 inverse formula in the theorem gives

$$A^{-1} = -\frac{1}{2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}.$$

Therefore,

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} -6 \\ 5 \end{bmatrix}$$

Check this solution by elimination in The Elimination Scratchpad

Activity: One inverse, several right-hand sides (U2-LO5)

Suppose that

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Use this information to calculate the solution to the following systems:

$$A\mathbf{x} = \mathbf{c}$$

$$\mathbf{x} = A^{-1}\mathbf{c} = \begin{bmatrix} -4 \\ 7 \\ \frac{7}{2} \end{bmatrix}$$

Check this solution by elimination in The Elimination Scratchpad

Activity: One inverse, several right-hand sides (U2-LO5)

Suppose that

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Use this information to calculate the solution to the following systems:

$$A\mathbf{x} = \mathbf{0}$$

$$\mathbf{x} = A^{-1}\mathbf{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Check this solution by elimination in The Elimination Scratchpad

Activity: A coordinate map and invertibility (U2-LO5)

Consider the transformation $P : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$P\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}\right) = \begin{bmatrix} a_1 \\ 0 \end{bmatrix}.$$

Is P a matrix transformation? If so, find the matrix A corresponding to P .

Yes. The map P preserves addition and scalar multiplication, so it is linear. By Every linear map has a matrix, its matrix has the images of the standard basis vectors as columns:

$$A = [P(\mathbf{e}_1) \ P(\mathbf{e}_2)] = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

To verify:

$$P\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} a_1 \\ 0 \end{bmatrix}$$

Activity: A coordinate map and invertibility (U2-LO5)

Consider the transformation $P : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

Activity: A coordinate map and invertibility (U2-LO5) (continued)

$$P\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}\right) = \begin{bmatrix} a_1 \\ 0 \end{bmatrix}.$$

Is P an invertible transformation?

No, P is not invertible. Note that $\det(A) = 0$, and so A is not an invertible matrix, and so P cannot be an invertible transformation.

Consider vectors $\mathbf{c}_1, \dots, \mathbf{c}_n$ in $\mathbb{R}^m$, and let A be the $m \times n$ matrix whose columns are the vectors $\mathbf{c}_1, \dots, \mathbf{c}_n$. Recall that $\{\mathbf{c}_1, \dots, \mathbf{c}_n\}$ are linearly independent precisely when the equation $A\mathbf{x} = \mathbf{0}$ has *only* the *trivial* solution (see Linear Independence and the uniqueness theorem, the theorem). By Spanning and solvability, $\text{span}\{\mathbf{c}_1, \dots, \mathbf{c}_n\} = \mathbb{R}^m$ exactly when $A\mathbf{x} = \mathbf{b}$ has a solution for all $\mathbf{b}$ in $\mathbb{R}^m$.

Theorem: Invertibility Criteria

Let A be an $n \times n$ matrix. The following are equivalent:

1. A is nonsingular/invertible.
2. The columns of A are linearly independent.
3. The columns of A span $\mathbb{R}^n$.
4. The rows of A are linearly independent.
5. The rows of A span $\mathbb{R}^n$.
6. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
7. A is row equivalent to I_n .

8. The linear system $Ax = b$ has a unique solution for every $n \times 1$ matrix b .
9. The rank of A is n .

Activity: Testing dependence using a matrix (U2-L04)

Let $S = \{v_1, v_2, v_3\}$, where

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} \quad v_3 = \begin{bmatrix} 3 \\ 0 \\ -5 \end{bmatrix}.$$

Is S linearly dependent?

Put the vectors in the columns of a matrix:

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & 0 \\ 2 & 2 & -5 \end{bmatrix}$$

Practice the homogeneous system in The Elimination Scratchpad Use c_1, c_2, c_3 for the coefficients in $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$.

Following the Gaussian elimination algorithm, we scale each pivot to 1 and eliminate the entries below it:

$$\begin{aligned}
& \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & 0 \\ 2 & 2 & -5 \end{bmatrix} \\
& \xrightarrow{\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 2R_1}} \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & -6 \\ 0 & 2 & -11 \end{bmatrix} \\
& \xrightarrow{R_2 \rightarrow -R_2} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 6 \\ 0 & 2 & -11 \end{bmatrix} \\
& \xrightarrow{R_3 \rightarrow R_3 - 2R_2} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -23 \end{bmatrix} \\
& \xrightarrow{R_3 \rightarrow -\frac{1}{23}R_3} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & 1 \end{bmatrix}.
\end{aligned}$$

This echelon form has a pivot in every column. Therefore the equation $Ac = 0$ has only the trivial solution, so no nontrivial linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ gives zero. Thus S is linearly independent, not linearly dependent.